

FMMVCC: Fuzzy Mamba-based Multi-View Contrastive Clustering for Univariate Time Series

1st Donato Cerciello 

*Departamento de Sistemas Informáticos,
Universidad Politécnica de Madrid
Madrid, Spain
donato.cerciello@upm.es*

2nd Leonardo Schiavo 

*Departamento de Sistemas Informáticos,
Universidad Politécnica de Madrid
Madrid, Spain
leonardo.schiavo@upm.es*

3rd Angel Panizo-LLedot 

*Departamento de Sistemas Informáticos,
Universidad Politécnica de Madrid
Madrid, Spain
angel.panizo@upm.es*

4th Javier Huertas Tato 

*Departamento de Sistemas Informáticos,
Universidad Politécnica de Madrid
Madrid, Spain
javier.huertas.tato@upm.es*

5th David Camacho 

*Departamento de Sistemas Informáticos,
Universidad Politécnica de Madrid
Madrid, Spain
david.camacho@upm.es*

Abstract—In many realistic scenarios, large volumes of time series data are generated with limited or expensive annotations. This limitation makes supervised learning methods difficult to apply and leads to the use of unsupervised approaches capable of discovering meaningful structures directly from raw data. Clustering therefore plays a crucial role in organizing time series into groups that share similar temporal patterns, enabling exploratory analysis and downstream tasks without requiring manual labeling. However, existing deep clustering methods often struggle to capture long-range temporal dependencies or rely on architectures with high computational cost. This paper introduces FMMVCC, a Mamba-based deep clustering framework for time series that leverages state space sequence modeling to efficiently learn temporal representations with linear complexity. Additionally, it utilizes multi-view self-supervised learning with temporal masking and augmentations. Experimental evaluation in 15 benchmark datasets proves that FMMVCC consistently outperforms state-of-the-art baselines, achieving the best overall performance in 29 of 60 total metric evaluations and the highest average rank in all tested scenarios.

Index Terms—Time Series Clustering, Contrastive Learning, Mamba SSM, Time Series Analysis.

I. INTRODUCTION

The rapid expansion of devices and sensor networks on the Internet of Things (IoT) has led to an explosion of time series data in domains such as smart cities, industrial systems, healthcare and environmental monitoring [1]. These continuous streams of temporal data contain valuable information for understanding patterns, detecting anomalies, and monitoring system behavior. However, obtaining labeled data in such environments is often expensive, time-consuming, or even infeasible due to the scale and complexity of sensor deployments. As a result, unsupervised learning methods, and in particular time series clustering, have become increasingly important for discovering latent structures in sequential data [2].

Time series clustering is defined as an unsupervised learning method that groups similar time series patterns without any

supervision. Traditional clustering methods, such as K-means and hierarchical clustering, have been shown to usually require handcrafted measures of similarity, such as Euclidean distance and DTW [3]. Although these methods have been shown to be successful in some cases, they have some difficulties in dealing with the complex temporal relationships and nonlinear dynamics that can be observed in recent high-dimensional time series data. For that reason, recent research has proposed deep learning-based clustering methods.

In recent years, contrastive learning (CL) has been established as an effective framework for the self-supervised representation learning method. By maximizing the agreement between different views of the same data and minimizing the agreement between different samples, CL has the ability to learn discriminative representations without the need for any supervision [4]. Several works have successfully applied CL to time series analysis by generating multiple augmented views of a signal through transformations. This enables the model to capture invariant representations that preserve the underlying temporal structure [?]. However, the design of effective multi-view strategies and the incorporation of clustering objectives is an area of research that needs to be addressed.

Another key challenge in time series modeling is capturing long-range temporal dependencies efficiently. Traditional recurrent architectures have been widely used for sequential modeling, but they often struggle with very long sequences and limited parallelization [5]. More recently, transformer-based architectures have demonstrated strong performance in sequence modeling tasks [6], yet their quadratic complexity with respect to sequence length can become computationally expensive in large-scale IoT environments. To address these limitations, state space models (SSMs) have recently re-emerged as an efficient alternative for sequence modeling. In particular, the Mamba architecture introduces a selective state space mechanism that enables linear-time sequence processing while maintaining strong modeling capacity for long temporal

contexts [7]. Despite its promising performance in sequence modeling tasks, its potential for unsupervised time series clustering remains largely unexplored [5].

To address these challenges, we propose Fuzzy Mamba-based Multi-View Contrastive Clustering (FMMVCC), a deep clustering framework for time series data. The proposed approach employs Mamba-based encoders to effectively capture long-range temporal dependencies while maintaining computational efficiency. To improve representation learning, multiple views of each time series are generated through temporal masking and augmentation strategies, enabling contrastive self-supervised training. In addition, a fuzzy clustering objective is introduced to produce soft cluster assignments, allowing the model to better handle ambiguous temporal patterns and improve cluster separability in the learned latent space.

A. Contributions

The main contributions of this work can be summarized as follows:

- **Mamba-based Representation Learning for Time Series Clustering.** A deep clustering framework is introduced that leverages Mamba-based encoders to model long-range temporal dependencies in time series data while maintaining linear-time computational complexity. This architecture provides an efficient alternative to recurrent and transformer-based models for large-scale sequential data.
- **Multi-View Masked Representation Learning.** Multiple views of each time series are generated through stochastic temporal augmentations combined with random temporal masking. The resulting views provide diverse but consistent observations of the same sequence, allowing the model to learn robust representations under both temporal perturbations and missing data.
- **Cluster-Oriented Representation Learning.** A clustering objective is integrated into the representation learning process to progressively structure the latent space and improve the separability of the cluster for time series data.

The paper is organized as follows. Section II reviews related work on deep clustering and time series representation learning. Section III describes the proposed framework. Section IV presents the experimental results. Lastly, section V discusses concluding remarks.

II. RELATED WORKS

A. Contrastive Learning for Time Series Clustering

CL has emerged as a powerful paradigm for self-supervised representation learning in time series, which enables the model to extract discriminative features without requiring any labels. The core idea is to maximize the agreement between different views of the same sample while minimizing similarity with other samples [4]. A significant number of recent studies show that the effectiveness of contrastive learning is largely dependent on the quality and variety of the views.

Current methods, such as TS2Vec [8], make use of hierarchical temporal contrasting to extract both local and global temporal

dependencies. Similarly, many methods make use of data augmentation methods such as scaling, jittering, and permutation to make the model invariant to common distortions in time series data [?]. However, these methods are mainly based on deterministic data enhancement and may not be effective in capturing uncertainty caused by missing data.

To address uncertainty in clustering, recent work has combined contrastive learning with fuzzy clustering. In particular, FCACC [9] introduces a cluster-aware contrastive framework with soft assignments, improving robustness in ambiguous time series scenarios.

In addition, the concept of temporal masking has also been explored as an effective approach to generate informative and challenging views. In these methods, the model is not required to reconstruct the input data, but to make it invariant to masking. This provides an opportunity for the model to extract robust and invariant representations even in incomplete data. In this direction, the EMTC [10] framework adapts the concept of evolving masking to effectively detect and reduce the relevance of less important timestamp. However, such approaches are typically built on attention-based architectures, which suffer from quadratic complexity and limited scalability for long sequences [6].

In contrast to existing approaches, the proposed framework combines contrastive learning with stochastic multi-view masking and cross-view reconstruction, enabling robust representation learning under missing data. It leverages the causal and selective dynamics of Mamba, allowing the model to capture temporal dependencies while filtering irrelevant information in a scalable manner.

B. Mamba-based Sequence Modeling for Time Series

Recently, State Space Models (SSMs) have been rediscovered as a promising solution, providing a linear complexity solution along with robust sequence modeling. Mamba provides a selective state space mechanism which filters relevant information based on input-dependent transitions. This enables efficient representation [7].

Recently, a study on PG-Mamba [11] was conducted to explore the use of Mamba-based clustering frameworks by incorporating a graph-based structure to capture global relationships between time series. Although such frameworks provide promising results, robustness is not a key focus of such frameworks.

In the presented framework, Mamba has been used within a multi-view contrastive setting to provide robustness via random masking and augmentation. A fuzzy objective is used to provide clustering. This enables efficient representation of long-range relationships within a representation that is invariant and discriminative.

III. METHODOLOGY

A complete overview of the FMMVCC framework is presented in Figure 1.

A. Multi-view Generation via Temporal Masking

Given an input time series $x \in \mathbb{R}^{T \times D}$, where T denotes the sequence length and D the number of variables, multiple views of the same signal are generated by stochastic augmentation and temporal masking. This strategy produces partially observed and perturbed versions of the original signal, simulating realistic conditions such as noisy or missing sensor measurements in IoT environments.

For each input time series, N views are generated as

$$x^{(v)} = M_v(A_v(x)), \quad v = 1, \dots, N$$

where $A_v(\cdot)$ denotes a stochastic augmentation operator and $M_v(\cdot)$ represents a masking operator.

The augmentation operator generates diverse views of the same signal through transformations commonly used in time series representation learning, including additive jittering, feature-wise scaling, and temporal permutation of subsequences. These transformations introduce controlled perturbations while preserving the underlying temporal structure. The masking operator simulates realistic missingness patterns by randomly removing both contiguous temporal segments and scattered observations across the sequence.

Each resulting masked view is processed independently by a dedicated encoder. This multi-view generation strategy encourages the model to learn representations that are robust to noise, missing observations, and temporal perturbations.

B. Mamba-Based Representation Learning

Each masked view is processed by a Mamba-based encoder that maps the input sequence into a latent representation space. Given a masked sequence $x^{(v)}$, the encoder produces a latent sequence representation

$$z^{(v)} = f_{\theta_v}(x^{(v)}), \quad z^{(v)} \in \mathbb{R}^{T \times d}$$

where d denotes the latent dimensionality. The encoder consists of an input projection layer followed by a stack of Mamba blocks and a linear output projection. Each Mamba block implements a selective state space model capable of modeling long-range temporal dependencies while maintaining linear-time computational complexity.

C. Cross-View and Intra-View Reconstruction

To enforce representation consistency, two types of decoder networks are introduced: intra-view decoders and cross-view decoders.

The intra-view decoder reconstructs the original input sequence from its latent representation:

$$\hat{x}^{(v)} = g_v(z^{(v)})$$

where $g_v(\cdot)$ denotes the decoder associated with view v . The reconstruction loss is defined as the mean squared error (MSE):

$$\mathcal{L}_{rec} = \frac{1}{V} \sum_{v=1}^V \|x - \hat{x}^{(v)}\|_2^2$$

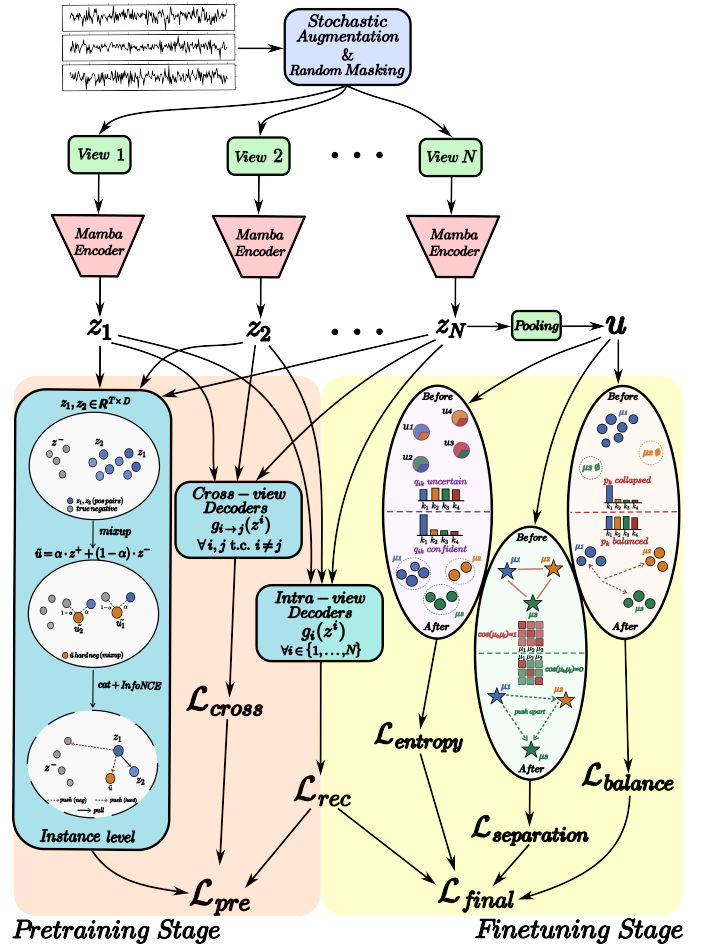


Fig. 1. Overview of the proposed FMMVCC framework. Multiple masked views of the input time series are encoded via Mamba-based encoders. During pretraining, intra-view and cross-view decoders enforce reconstruction and consistency, and a contrastive objective aligns representations across views. In the finetuning stage, the fused representations are optimized with a clustering objective to produce soft cluster assignments.

In addition, cross-view decoders enforce consistency between representations of different views. For two views i and j , the latent representation from one view is used to reconstruct the latent representation of the other view:

$$\tilde{z}^{(j)} = g_{i \rightarrow j}(z^{(i)})$$

The cross-view reconstruction loss is defined as

$$\mathcal{L}_{cross} = \frac{1}{V(V-1)} \sum_{i \neq j} \|z^{(j)} - \tilde{z}^{(j)}\|_2^2$$

To further align the representations obtained from different views, an instance-level contrastive objective is applied to the reconstructed latent representations. Given two reconstructed representations $\tilde{z}^{(i)}$ and $\tilde{z}^{(j)}$ corresponding to the same time series, the model maximizes their similarity while contrasting them against representations of other samples in the batch.

Formally, the contrastive loss is defined as

$$\mathcal{L}_{contrast} = -\log \frac{\exp(\text{sim}(\tilde{z}_i, \tilde{z}_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(\tilde{z}_i, \tilde{z}_k)/\tau)}$$

where $\text{sim}(\cdot, \cdot)$ denotes a similarity function, τ is a temperature parameter, and N denotes the number of samples in the batch. Additionally, a mixup-based strategy is employed to generate harder negative samples by combining positive and negative representations, improving the discriminative capability of the learned latent space.

The overall pretraining objective combines all the previous losses:

$$\mathcal{L}_{pre} = \lambda_1 \mathcal{L}_{contrast} + \lambda_2 \mathcal{L}_{cross} + \lambda_3 \mathcal{L}_{rec}$$

where λ_1 , λ_2 , and λ_3 control the contribution of each component.

D. Clustering-Oriented Finetuning

After the pretraining stage, the model is finetuned using a clustering-oriented objective designed to structure the latent space and improve cluster separability.

Given the latent sequence representations $\{z^{(v)}\}_{v=1}^V$, a temporal pooling mechanism aggregates each sequence into a fixed-length vector, which is then averaged across views to obtain a single representation $u \in \mathbb{R}^d$ per sample.

Soft cluster assignments are computed through a cosine-similarity-based softmax over a set of learnable cluster prototypes $\{\mu_k\}_{k=1}^K$:

$$q_{ik} = \frac{\exp(\langle \bar{u}_i, \bar{\mu}_k \rangle / \tau)}{\sum_{j=1}^K \exp(\langle \bar{u}_i, \bar{\mu}_j \rangle / \tau)}$$

where $\bar{u}_i$ and $\bar{\mu}_k$ denote L_2 -normalized representations and prototypes, and τ is a temperature parameter.

The clustering objective combines three complementary terms:

$$\begin{cases} \mathcal{L}_{entropy} = \frac{1}{N} \sum_{i,k} q_{ik} \log q_{ik} \\ \mathcal{L}_{balance} = \sum_k p_k \log p_k, \quad p_k = \frac{1}{N} \sum_i q_{ik} \\ \mathcal{L}_{separation} = \frac{1}{K(K-1)} \sum_{i \neq j} (\bar{\mu}_i^\top \bar{\mu}_j)^2 \end{cases}$$

The entropy term encourages confident assignments, the balance term mitigates cluster collapse, and the separation term promotes diversity among prototypes, yielding the clustering objective

$$\mathcal{L}_{cluster} = \mathcal{L}_{entropy} + \lambda_b \mathcal{L}_{balance} + \lambda_s \mathcal{L}_{separation}$$

where λ_b and λ_s are respectively the balance and the separation weights. The overall finetuning objective is therefore defined as

$$\mathcal{L}_{final} = \lambda_r \mathcal{L}_{rec} + \lambda_c \mathcal{L}_{cluster}$$

where λ_r and λ_c control the contribution of each component.

TABLE I
UCR UNIVARIATE TIME-SERIES DATASETS USED IN EXPERIMENTS

Dataset	Samples	Length	Classes
ACSF1	100	1460	10
Adiac	390	176	37
AllGestureWiimoteX	300	500	10
AllGestureWiimoteZ	300	500	10
BirdChicken	20	512	2
Car	60	577	4
CricketX	390	300	12
CricketZ	390	300	12
Crop	7200	46	24
DistalPhalanx	400	80	3
ECG200	100	96	2
ElectricDevices	8926	96	7
FaceAll	560	131	14
FreezerRegularTrain	150	301	2
SyntheticControl	300	60	6

IV. EXPERIMENTAL RESULTS

A. Dataset

The proposed method is evaluated on datasets from the UCR Time Series Archive [12], a widely used benchmark in the time series data mining community. The archive contains a large collection of univariate time series datasets spanning various domains. As described in Table I, a subset of representative datasets has been selected to assess the performance of clustering under varying sequence lengths and data characteristics.

B. Comparison Methods

The proposed approach has been compared against a set of representative baselines for time series clustering.

- **TimeSeriesKMeans-DTW (TSK-DTW)** [13] is a classical clustering method that extends K-means using Dynamic Time Warping as similarity measure, providing a strong non-deep baseline.
- **HDBSCAN** [14] is a hierarchical density-based clustering algorithm that extends DBSCAN by constructing a hierarchy of clusters, enabling robust clustering with varying densities and improved handling of noise.
- **FCACC** [9] is a deep clustering framework that integrates contrastive learning with fuzzy clustering, enabling soft cluster assignments in the learned latent space.
- **PG-Mamba** [11] is a recent model that combines Mamba-based sequence modeling with a graph-based clustering framework, leveraging relational structures between time series.
- **EMTC** [10] is a multi-view self-supervised approach that employs masking strategies and contrastive learning to learn robust representations for time series clustering.

C. Evaluation Metrics

The clustering performance is evaluated using the following standard metrics:

- **NMI (Normalized Mutual Information)**: measures the shared information between predicted clusters and ground-truth labels.

TABLE II
CLUSTERING PERFORMANCE COMPARISON ACROSS MULTIPLE DATASETS.
BEST RESULTS ARE HIGHLIGHTED IN RED AND SECOND-BEST IN BLUE.

Datasets	Metrics	Baselines					Proposed
		TSK-DTW	HDBSCAN	PG-Mamba	FCACC	EMTC	FMMVCC
ACSF1	F1	0.0621	0.0483	0.1191	0.2018	0.1516	0.1635
	NMI	0.3270	0.2084	0.3409	0.6287	0.3413	0.6227
	ARI	0.0787	0.0449	0.0966	0.7466	0.0854	0.7641
	RI	0.6166	0.4713	0.8420	0.8909	0.7598	0.8980
Adiac	F1	0.0353	0.0000	0.0315	0.0067	0.0226	0.0689
	NMI	0.6162	0.3043	0.3845	0.6859	0.5294	0.7134
	ARI	0.2292	0.0195	0.0294	0.9176	0.1317	0.9264
	RI	0.9452	0.3472	0.9359	0.9677	0.9497	0.9713
AllGestureWiimoteX	F1	0.0685	0.0510	0.0074	0.0546	0.0888	0.1643
	NMI	0.3577	0.0519	0.2404	0.4319	0.1468	0.4016
	ARI	0.2466	0.0041	0.0457	0.5557	0.0485	0.4501
	RI	0.8624	0.2975	0.8827	0.8696	0.7828	0.8250
AllGestureWiimoteZ	F1	0.0496	0.0610	0.0185	0.1395	0.0942	0.1654
	NMI	0.2892	0.1058	0.1181	0.3696	0.1028	0.2901
	ARI	0.1363	0.0419	0.0111	0.4720	0.0321	0.4417
	RI	0.8246	0.6657	0.8751	0.8361	0.8060	0.8224
BirdChicken	F1	0.5000	0.4486	0.1067	0.3484	0.4949	0.8880
	NMI	0.0000	0.0073	0.2982	0.0217	0.0000	0.6062
	ARI	0.0556	0.0444	0.0870	0.0605	0.0532	0.6858
	RI	0.4737	0.4789	0.5579	0.5500	0.4737	0.8577
Car	F1	0.3330	0.1343	0.1090	0.6081	0.1684	0.4166
	NMI	0.2542	0.1787	0.0535	0.5935	0.0910	0.2219
	ARI	0.1361	0.0877	0.0282	0.6727	0.0175	0.2845
	RI	0.6271	0.6644	0.6322	0.8408	0.5181	0.6417
CricketX	F1	0.2002	0.0403	0.0128	0.0324	0.1212	0.1526
	NMI	0.4642	0.0812	0.2403	0.4437	0.1212	0.3695
	ARI	0.2472	0.0207	0.0294	0.7701	0.0204	0.7460
	RI	0.8749	0.5700	0.8924	0.9026	0.8257	0.8913
CricketZ	F1	0.0709	0.0460	0.0236	0.0746	0.0668	0.1223
	NMI	0.4557	0.0997	0.2450	0.4903	0.1568	0.4632
	ARI	0.2339	0.0320	0.0424	0.7778	0.0402	0.7821
	RI	0.8778	0.6671	0.8874	0.9060	0.8366	0.9082
Crop	F1	0.0388	0.0044	0.0003	0.0228	0.0387	0.0474
	NMI	0.4101	0.3851	0.2841	0.5367	0.4110	0.5374
	ARI	0.2295	0.1547	0.0172	0.6781	0.1816	0.7096
	RI	0.9271	0.8087	0.9568	0.9252	0.9308	0.9335
DistalPhalanx	F1	0.3016	0.0906	0.0090	0.3824	0.1637	0.4335
	NMI	0.3647	0.2418	0.1551	0.3313	0.1277	0.4556
	ARI	0.2625	0.1983	0.0842	0.4485	0.1188	0.6231
	RI	0.6537	0.5878	0.6102	0.7586	0.5812	0.8286
ECG200	F1	0.5062	0.7455	0.0043	0.3850	0.3200	0.8354
	NMI	0.0044	0.1885	0.1304	0.0498	0.0707	0.3635
	ARI	0.0170	0.2791	0.0397	0.1147	0.1166	0.4950
	RI	0.5240	0.6422	0.4933	0.5739	0.5604	0.7508
ElectricDevices	F1	0.1145	0.1229	0.0227	0.1543	0.1026	0.2070
	NMI	0.3952	0.1421	0.0000	0.5630	0.1579	0.3887
	ARI	0.3169	0.0456	0.0000	0.7791	0.0770	0.5721
	RI	0.7979	0.4529	0.1774	0.8987	0.7306	0.7860
FaceAll	F1	0.0820	0.0413	0.0043	0.0407	0.0892	0.1076
	NMI	0.6771	0.3615	0.1204	0.5181	0.1705	0.3063
	ARI	0.5839	0.1676	0.0090	0.4570	0.0683	0.4295
	RI	0.9281	0.7767	0.8925	0.8426	0.8372	0.8542
FreezerRegularTrain	F1	0.7592	0.0837	0.0009	0.2735	0.5714	0.7209
	NMI	0.2070	0.1430	0.1199	0.0887	0.0902	0.1726
	ARI	0.2694	0.1361	0.0160	0.1175	0.0544	0.1986
	RI	0.6347	0.5681	0.5082	0.5588	0.5271	0.5993
SyntheticControl	F1	0.1603	0.0303	0.0286	0.0738	0.0650	0.3153
	NMI	0.9475	0.6563	0.1059	0.7297	0.4849	0.4635
	ARI	0.9376	0.4811	0.0144	0.6795	0.3551	0.5933
	RI	0.9829	0.7971	0.7998	0.8530	0.8142	0.8117
Global	#Best	12	0	3	16	0	29
	#Second	8	5	5	23	2	17
	Avg Rank	2.98	4.68	4.67	2.47	4.35	1.85

- **RI (Rand Index):** evaluates the pairwise agreement between predicted and true cluster assignments.
- **ARI (Adjusted Rand Index):** measures the similarity between clusterings while correcting for chance.
- **F1-score:** represents the harmonic mean of precision and recall.

All metrics range between 0 and 1, where higher values indicate better clustering performance.

D. Experimental Details

For the baseline methods, we implement them using open-source code, following the parameter configurations and experimental setups specified in their respective papers.

For the proposed FMMVCC model, the specific parameter settings are $\lambda_b = 0.2$ and $\lambda_s = 0.5$. These values were selected based on a preliminary sensitivity analysis to ensure an optimal balance between cluster separation and assignment stability, as further described in Section IV-F. The batch size is set to 32, the latent representation dimension is fixed to 64 and the number of views N is equal to 4. The model is trained using the AdamW optimizer with an initial learning rate of 1e-3 during the pretraining phase and 1e-4 during the finetuning phase. The number of pretraining epochs and finetuning stage is set to 100.

All training and evaluation procedures were performed using a NVIDIA Quadro RTX 8000.

E. Performance Comparing

Table II reports the clustering results across all datasets, where the best performances are highlighted in red and the second-best in blue. In general, FMMVCC achieves the highest number of best scores (29) and the lowest average per-metric rankings across all datasets (1.85), demonstrating a clear and consistent advantage over all baselines.

This advantage is particularly evident in structure-aware metrics such as ARI and NMI, where FMMVCC often ranks first or second. This suggests that the learned latent space better captures the underlying cluster structure, rather than simply optimizing pairwise agreements (as reflected by RI). In contrast, some baselines achieve competitive RI values but significantly lower ARI, indicating less meaningful cluster assignments.

Compared to other deep learning approaches, FMMVCC shows a clear improvement in robustness and stability. Although methods such as FCACC achieve top performance in some cases, their results are more variable, as reflected by a higher average rank (2.47). Similarly, EMTC and PG-Mamba exhibit less consistent behavior across metrics, suggesting that masking or Mamba-based modeling alone is not sufficient without a tightly integrated multi-view contrastive framework. Traditional methods such as TSK-DTW and HDBSCAN generally obtain lower average rankings; however, they remain competitive in specific scenarios, occasionally achieving strong results, particularly in pairwise-based metrics or when the cluster structure is well aligned with the underlying similarity measure. This highlights that classical approaches can still be effective in simpler settings, but lack the representational flexibility required to consistently capture complex temporal dependencies across diverse datasets.

Overall, the results indicate that combining contrastive multi-view learning, temporal masking, and the causal selective dynamics of Mamba leads to a more stable and structurally consistent latent space, resulting in improved clustering performance across a wide range of scenarios.

F. Parameter Analysis

To evaluate the effect of fuzzy clustering loss components, a sensitivity analysis is conducted on the *SyntheticControl* dataset by varying the separation weight λ_s and the balance

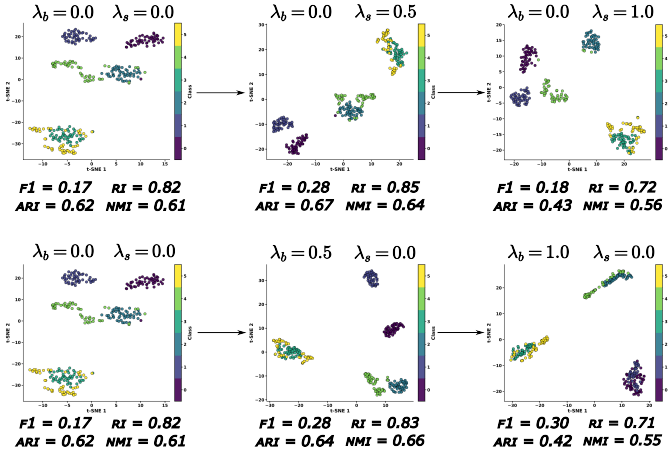


Fig. 2. Impact of loss weights on cluster formation. The top row depicts how the separation term λ_s enhances cluster isolation by maximizing inter-centroid distances. The bottom row demonstrates the influence of the balance term λ_b on cluster distribution uniformity.

TABLE III
PER-METRIC ANALYSIS OF BIDIRECTIONAL VS. UNIDIRECTIONAL MAMBA ENCODERS.

Avg Metric	Unidirectional	Bidirectional	Δ %
F1	0.3206	0.2873	↓ 9.41%
NMI	0.4251	0.3814	↓ 11.57%
ARI	0.5801	0.5245	↓ 12.04%
RI	0.8253	0.7954	↓ 3.64%

weight λ_b . Fig. 2 shows the corresponding t-SNE projections [15].

The first row illustrates the impact of $\lambda_s \in \{0, 0.5, 1.0\}$ with $\lambda_b = 0$. When $\lambda_s = 0$, latent representations exhibit partial overlap, making the boundaries of the cluster less distinguishable. As λ_s increases, clusters become progressively more separated, with clearer margins between groups. This indicates that the separation term effectively enhances the inter-cluster distance and improves the discriminability of the learned representations.

The second row shows the effect of $\lambda_b \in \{0, 0.5, 1.0\}$ with $\lambda_s = 0$. In this case, increasing λ_b leads to more compact and structured clusters, with samples concentrating more tightly around their respective centers. This suggests that the balance term encourages a more uniform utilization of clusters and reduces ambiguity in the assignments, resulting in lower intra-cluster variance.

Overall, the results highlight the complementary roles of the two components: λ_s primarily controls inter-cluster separation, while λ_b improves intra-cluster compactness.

G. Ablation Study

To assess the impact of the sequence modeling strategy, we compare the unidirectional Mamba encoder with a bidirectional variant, where forward and backward representations are combined through a learnable gate [16].

As shown in Table III, the bidirectional configuration con-

sistently degrades performance across all metrics. The largest drops are observed in ARI (−12.04%) and NMI (−11.57%), indicating a weaker alignment with the structure of the ground-truth, followed by F1 (−9.41%). RI shows a smaller decrease (−3.64%), but still confirms the overall trend.

This behavior suggests that enforcing a causal, unidirectional representation is more suitable for the proposed framework. In particular, the bidirectional setting may introduce temporal inconsistencies across masked views, reducing the effectiveness of the contrastive objective and leading to less structured latent representations.

V. CONCLUSIONS

This paper presents an innovative deep clustering framework for univariate time series, integrating Mamba with a multi-view contrastive learning strategy. Our model effectively captures long-range temporal dependencies by taking advantage of the Mamba architecture’s linear complexity. The addition of a stochastic temporal masking and augmentation strategy ensures that the learned representations can handle noise and missing data, which are common problems in real-world IoT and sensor environments. The addition of a fuzzy clustering objective also makes it possible to assign soft clusters, which makes it easier to deal with unclear temporal patterns.

Experimental evaluations conducted on several benchmark datasets from the UCR Archive show that FMMVCC consistently outperforms the baseline methods. Our analysis confirms that a unidirectional encoding strategy, in conjunction with inter-view and intra-view reconstruction losses, is optimal for preserving temporal consistency in a self-supervised framework. The parameter sensitivity study underscores the significance of the separation and balance terms in the configuration of a discriminative latent space.

Despite these results, the current framework is primarily designed for univariate sequences, and its latent representations lack direct interpretability. Therefore, future research will focus on extending the framework to multivariate time series to capture inter-variable correlations, alongside conducting a deeper latent space analysis to enhance the explainability of the learned temporal features and their influence on cluster formation.

DATA AND RESOURCES

The data used in this study are in UCR Time Series Classification Archive. The implementation of FMMVCC is available on <https://github.com/DonatoCerciello/FMMVCC>.

ACKNOWLEDGMENT

This work was supported by the MSCA Doctoral Networks project T.U.A.I. - Towards an Understanding of Artificial Intelligence via a transparent, open and explainable perspective (TUAI) project, N°101168344.

REFERENCES

- [1] U. Schlegel, G. M. Tavares, and T. Seidl, “Towards explainable deep clustering for time series data,” *arXiv preprint arXiv:2507.20840*, 2025.
- [2] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Jin, Y. Liu, J. Y. Zhang, Y. Liang, G. Pang, D. Song, and S. Pan, “Self-supervised learning for time series analysis: Taxonomy, progress, and prospects,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
- [3] J. Paparrizos, F. Yang, and H. Li, “Bridging the gap: A decade review of time-series clustering methods,” *arXiv preprint arXiv:2412.20582*, 2024.
- [4] Q. Meng, H. Qian, Y. Liu, Y. Xu, Z. Shen, and L. Cui, “Unsupervised representation learning for time series: A review,” *arXiv preprint arXiv:2308.01578*, 2023.
- [5] S. Somvanshi, M. M. Islam, M. S. Mimi, S. B. B. Pollock, G. Chhetri, and S. Das, “From s4 to mamba: A comprehensive survey on structured state space models,” *arXiv preprint arXiv:2503.18970*, 2025.
- [6] Q. Wen, T. Zhou, C. Zhang, W. Chen, Z. Ma, J. Yan, and L. Sun, “Transformers in time series: A survey,” *arXiv preprint arXiv:2202.07125*, 2022.
- [7] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *First conference on language modeling*, 2024.
- [8] Z. Yue *et al.*, “Ts2vec: Towards universal representation of time series,” in *AAAI*, 2022.
- [9] C. Wang, M. Du, X. Jiang, and Y. Dong, “Fuzzy cluster-aware contrastive clustering for time series,” *Pattern Recognition*, 2025.
- [10] Z. Tan, X. Luo, Y. Liu, and Y. Zhang, “Mask the redundancy: Evolving masking representation learning for multivariate time-series clustering,” *arXiv preprint arXiv:2511.17008*, 2025.
- [11] Y. Sun, D. Zuo, and J. Gao, “Pg-mamba: An enhanced graph framework for mamba-based time series clustering,” *Sensors*, vol. 25, no. 16, p. 5043, 2025.
- [12] H. A. Dau, A. Bagnall, K. Kamgar, C.-C. M. Yeh, Y. Zhu, S. Gharghabi, C. A. Ratanamahatana, and E. Keogh, “The ucr time series archive,” *IEEE/CAA Journal of Automatica Sinica*, vol. 6, no. 6, pp. 1293–1305, 2019.
- [13] F. Petitjean, A. Ketterlin, and P. Gançarski, “A global averaging method for dynamic time warping, with applications to clustering,” *Pattern recognition*, vol. 44, no. 3, pp. 678–693, 2011.
- [14] L. McInnes and J. Healy, “Accelerated hierarchical density based clustering,” in *2017 IEEE International Conference on Data Mining Workshops (ICDMW)*. IEEE, 2017.
- [15] M. Vlaić, I. Mikulić, G. Delač, M. Šilić, and K. Vladimir, “A review of time series dimensionality reduction methods,” in *2025 MIPRO 48th ICT and Electronics Convention*. IEEE, 2025, pp. 149–154.
- [16] A. Liang, X. Jiang, Y. Sun, X. Shi, and K. Li, “Bi-mamba+: Bidirectional mamba for time series forecasting,” *arXiv preprint arXiv:2404.15772*, 2024.